

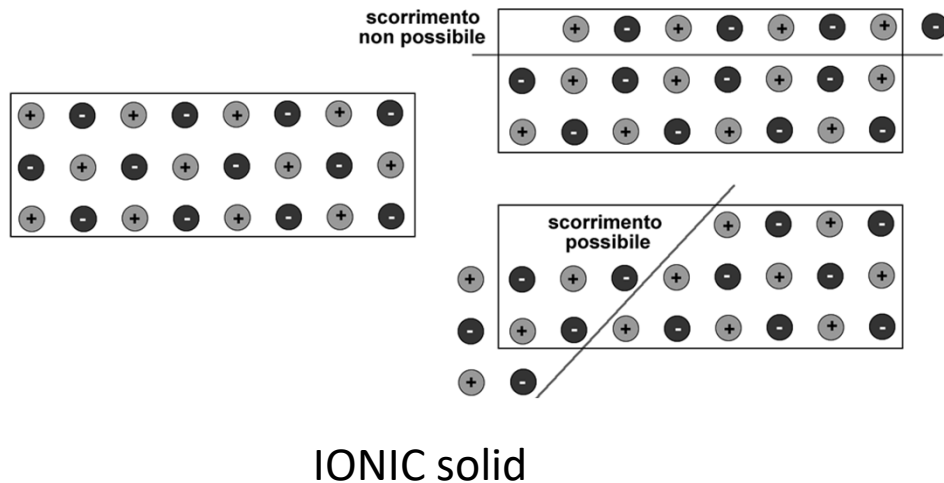
# **CERAMIC MATERIALS**

# General properties – the bonds

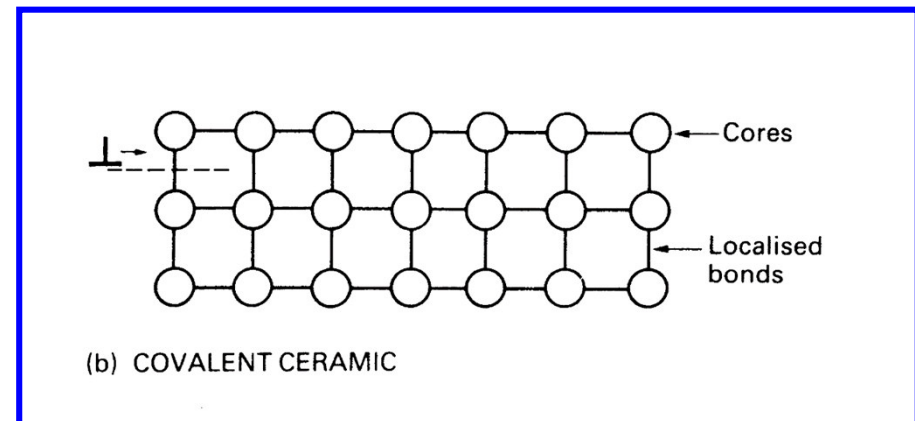
Most ceramics are compounds of non-metallic and metallic elements for which the interatomic bonds can be totally ionic, totally covalent, or mixed.

- Energetically very strong directional bonds
- No free electrons

Ceramic (“keramos”) = “burnt stuff”, indicating the method to produce these desired properties



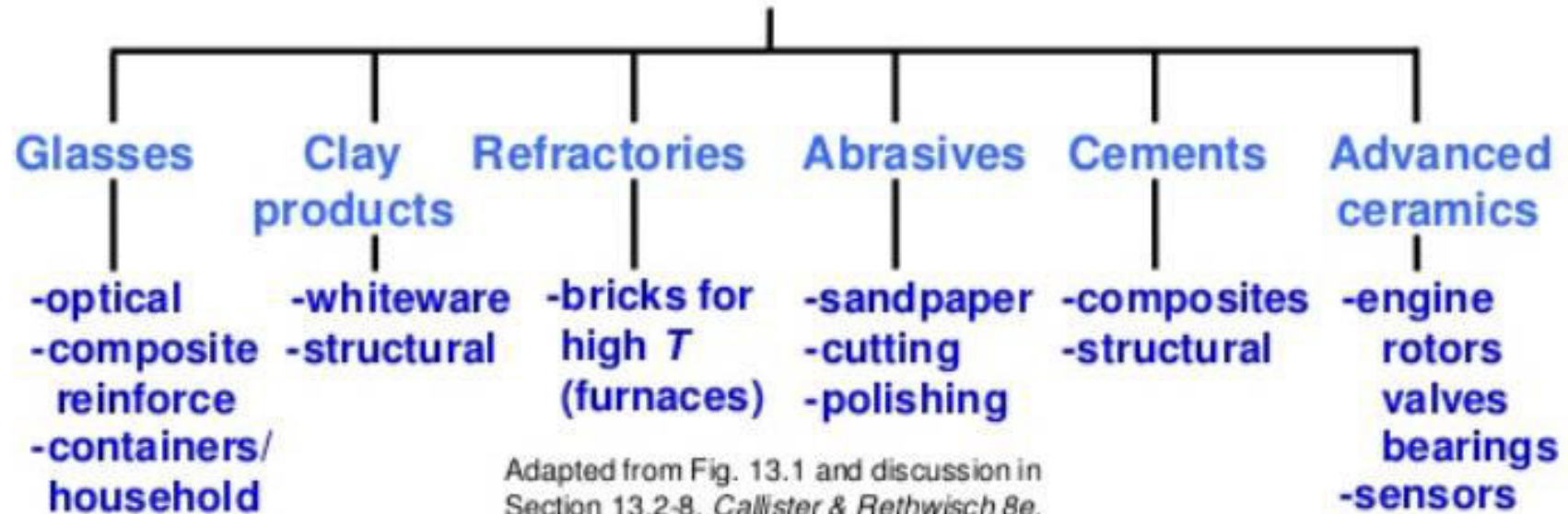
COVALENT solid



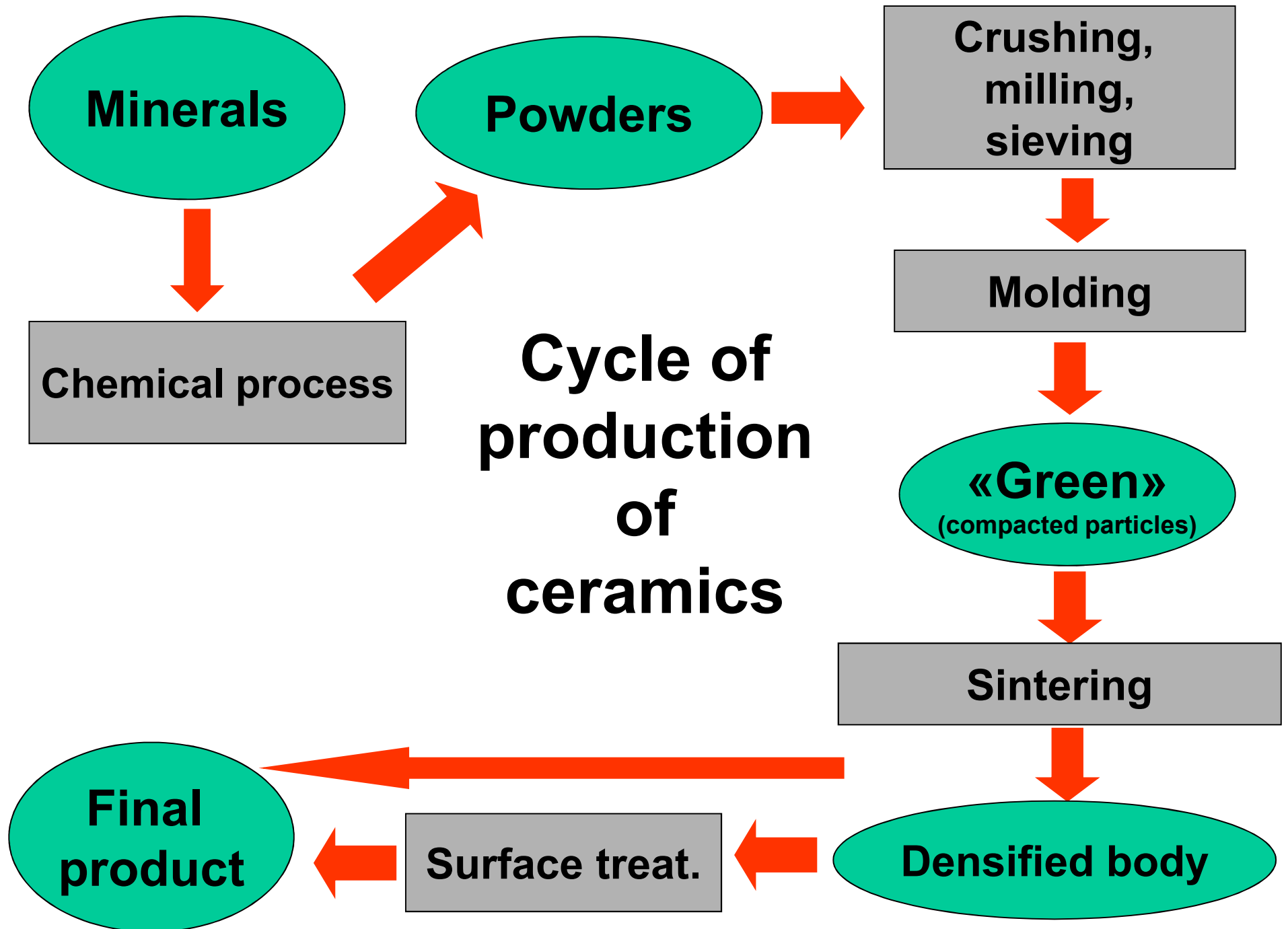
# Main characteristics

- Good compressive strength
- Very poor tensile and bending strength
- Fragility (low toughness)
- High hardness
- Good heat resistance (**refractoriness**)
- Good thermal and electrical insulation (brick thermal conductivity = 0.15-0.8 W / mK, porcelain 2 W / mK)
- Good chemical inertia (resistance to acid and alkaline attacks)
- Poor resistance to thermal shock

# Ceramic Materials



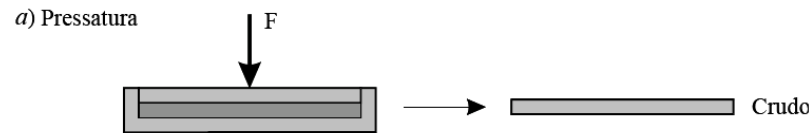
Adapted from Fig. 13.1 and discussion in Section 13.2-8, *Callister & Rethwisch 8e*.



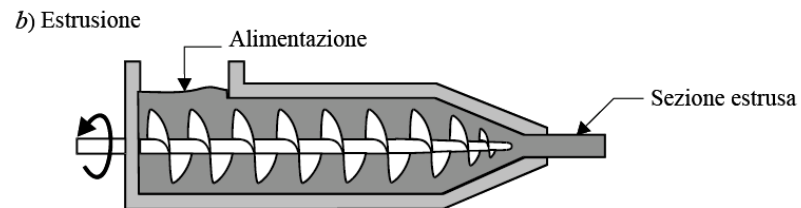
# Molding methods

Three common techniques used to manufacture clay-based materials:

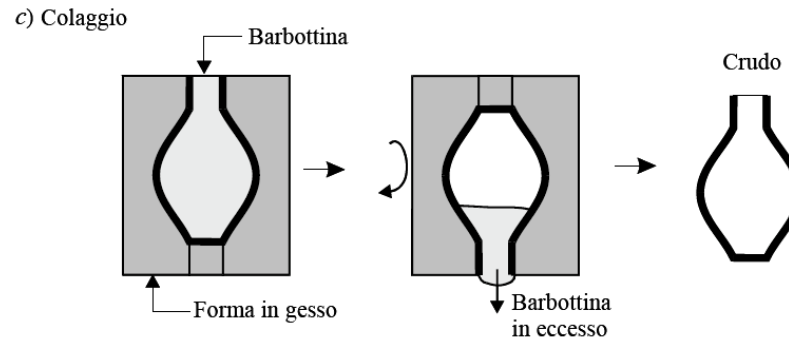
1) powder pressing



2) Extrusion



3) slip casting

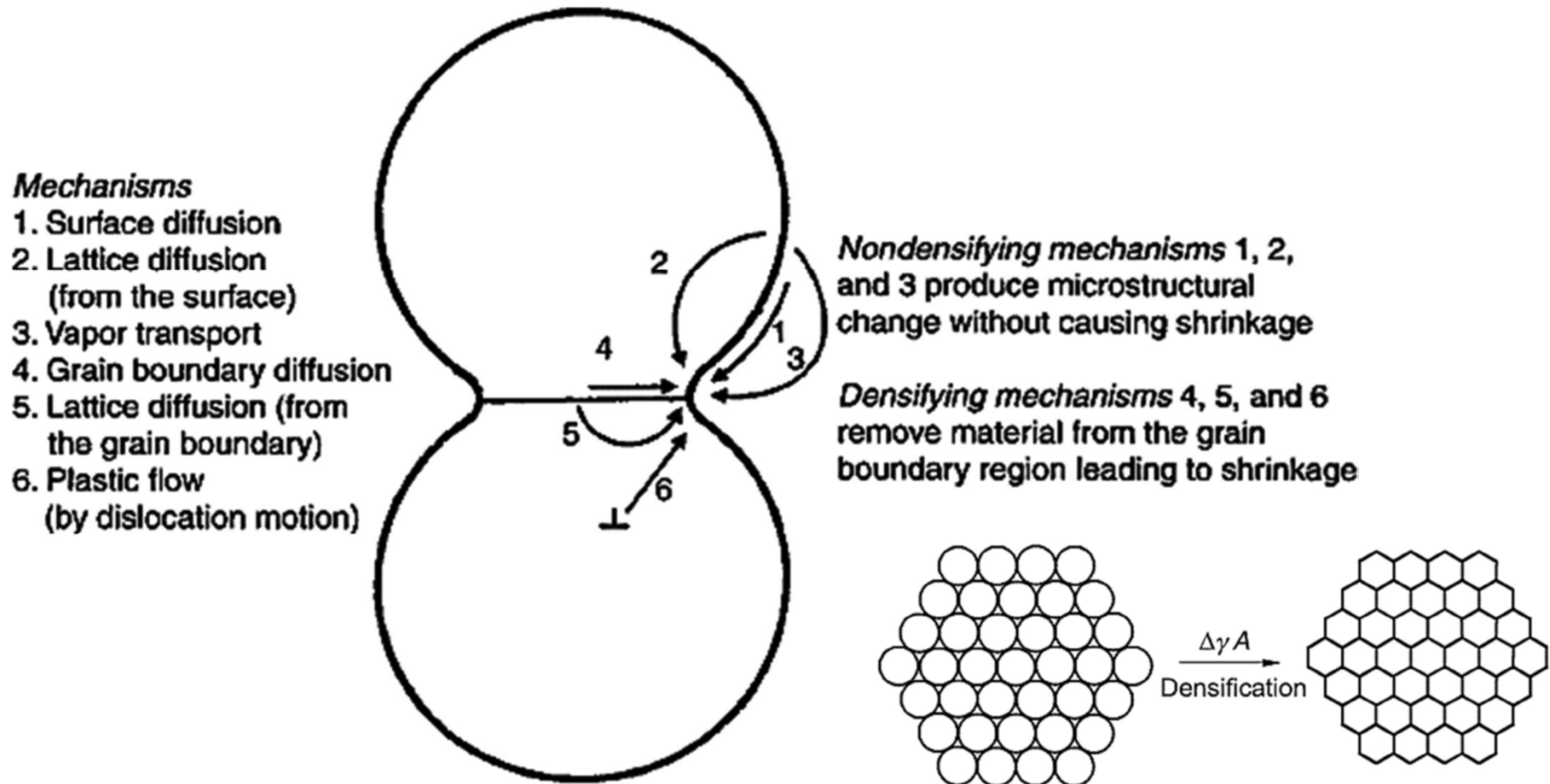


| Method          | Water in the paste (%) |
|-----------------|------------------------|
| Powder pressing | 5-7                    |
| Extrusion       | 15-20                  |
| Slip casting    | 35-40                  |

# Sintering

The temperature used for firing depends on the composition and desired properties to achieve.

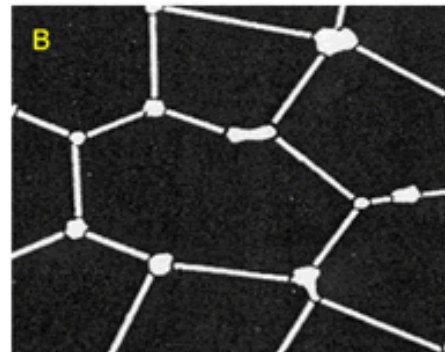
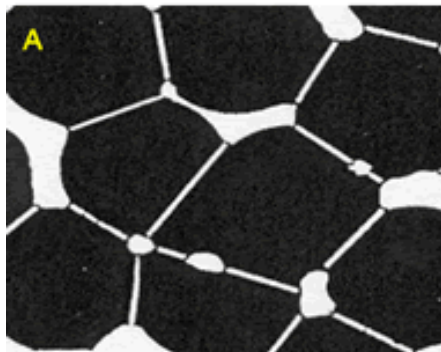
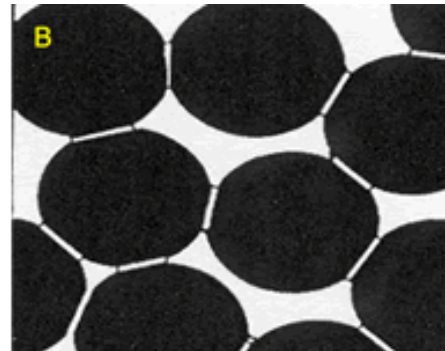
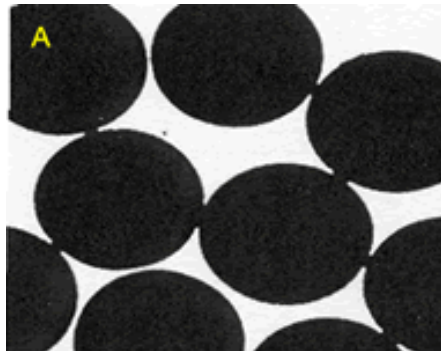
For traditional ceramics - clay-based materials/bricks – the sintering temperature is above 1000 °C.



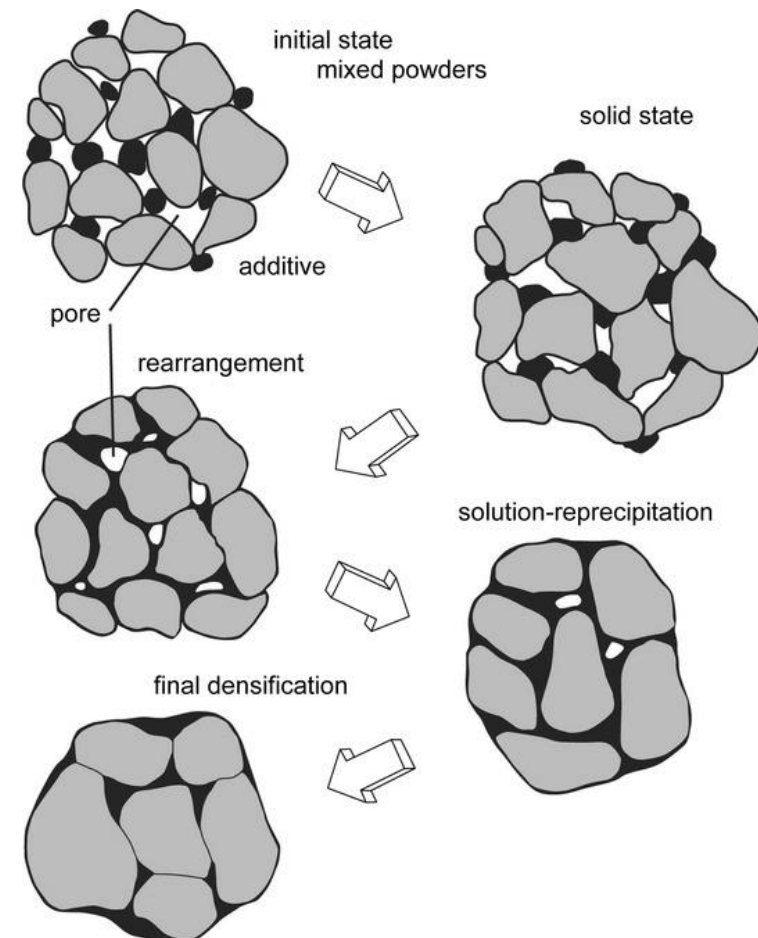
**Use of sintering aids → vitrification:** formation of a liquid glass that flows into and fills the pore volume.

Overall effect: acceleration of the atomic diffusion (rate > compared to solid state sintering)

*Solid state sintering*



*vs. use of sintering aids*





# Shrinkage due to sintering

